

Response to the Editor and Reviewers

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Manuscript Type: Regular Paper

Manuscript: *Shielded Q-learning for Thermally-Safe, Energy-Aware Multi-Model Intrusion Detection Scheduling on Lightweight Edge Devices*

Target Journal: IEEE Transactions on Sustainable Computing

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Handling Editor: Prof. Ahmed Louri, Editor-in-Chief; Associate Editor

How to Read This Letter

- **Blue boxes** reproduce the substantive Associate Editor and Reviewer comments.
- **Green boxes** contain our point-by-point responses and revision actions.
- **Orange boxes** identify the corresponding locations in the revised manuscript.
- **Purple boxes** show representative revised text or interpretation changes for quick verification.

Author Response

Dear Editor-in-Chief, Associate Editor, and Reviewers,

We sincerely thank the Associate Editor and all reviewers for the detailed and constructive comments. We have revised the manuscript substantially while retaining the submitted paper's title, FSSQL-R method, 40-ms latency cap, 82°C thermal cap, and 100-ms decision-window formulation.

Because the earlier experimental artifacts were no longer available, the empirical implementation for this revision was rebuilt from the submitted method definitions and frozen before evaluation. The revised detector, preprocessing, calibration, and scheduling artifacts are therefore not claimed to be byte-identical reproductions of the historical experimental artifacts. Accordingly, the revised tables replace, rather than numerically reproduce, the earlier replay tables.

The submitted manuscript also stated that test-set TPR/FPR supplied detector-quality terms to the scheduler reward. As written, this allowed test information to influence the controller objective. We treat this as a methodological leakage path and have eliminated it: the revision fixes all reward-quality terms from the validation partition only, while the official 82,332-row UNSW-NB15 test set remained sealed until all development, calibration, hardware, and statistical decisions were frozen and was then opened once for final detector-generalization evaluation.

Summary of Major Revisions

Author Response

- C1. Rebuilt deployment evidence.** Replaced the earlier replay-centered deployment evidence with independently calibrated online hardware evaluation on Raspberry Pi 3B+ and Raspberry Pi 4B 8GB.
- C2. Added a strong no-training baseline.** Added the conventional Threshold controller requested by the reviewers; it is the strongest detector-selection baseline by mean F1

on the Pi 4B conditions.

- C3. Added paired-seed inference and learning-stability analysis.** Reported seed-level dispersion, bootstrap intervals, paired tests, effect sizes, state/action-space size, exploration/learning schedules, and 600-to-6000-window stability diagnostics.
- C4. Measured complete online deployment cost.** Instrumented preprocessing, telemetry, state construction, guard evaluation, action selection, Q lookup, detector inference, switching, controller time, and complete E2E latency.
- C5. Added physical energy measurements.** Measured Pi 4B DC input-side power and energy with a POWER-Z KM003C bound one-to-one to 60 accepted online runs; the data do not establish a consistent FSSQL-R energy advantage.
- C6. Recalibrated safety evidence per device.** Used independent fit/margin/coverage splits on both Pi platforms and report empirical held-out coverage rather than an unconditional end-to-end hard-safety claim.
- C7. Added cross-device and background-load robustness.** Evaluated a second Raspberry Pi platform and B0/B1/B2 background conditions without reusing Pi 3B+ guard parameters.
- C8. Corrected test-information leakage.** Scheduler reward quality terms now come only from the validation partition; the official test is used once, after all choices are frozen.
- C9. Calibrated the interpretation of Q-learning.** FSSQL-R is not presented as universally superior: it underperforms Safe-Greedy in the Pi 3B+ stationary binding regime, while showing higher F1/reward than that same myopic reference on the Pi 4B; Threshold remains stronger by mean F1 and FPR.

Revised Text – Representative Excerpt

Key revised interpretation.

The revised evidence supports calibrated feasible-set enforcement, but not universal superiority of the learning layer. The incremental value of Q-learning is device- and regime-dependent, and direct physical-energy measurements do not establish a consistent FSSQL-R energy advantage.

Associate Editor

Reviewer / AE Comment

The contribution needs to be separated more clearly from prior work and simpler controllers; clarify the relation to earlier energy-aware/multi-objective Q schedulers, discuss recent shielded-RL IDS work, clarify Unshielded-Q, and add a conventional threshold controller.

Author Response

We agree. The revised Related Work explicitly states that shielding and constrained RL are established and does not claim novelty for shielding itself. We added the 2026 paper by Malwi *et al.*, “Safety-Aware Reinforcement Learning for Self-Healing Intrusion Detection in 5G-Enabled IoT Networks,” DOI 10.32604/cmc.2026.074274, and distinguish its self-

healing remediation actions from our model-selection/physical-resource setting. We also substantially expanded the recent literature review with 2024–2025 TSUSC work on DRL power/frequency control, energy-harvesting IoT scheduling, latency–energy task scheduling, DVFS for edge workloads, and IoT anomaly adaptation, together with recent TMC work on workload-adaptive edge inference/DVFS and TIFS work on online and open-set IDS. A new comparison table makes the difference in action semantics, measurement boundary, and safety treatment explicit. We also clarify that Unshielded-Q is the revised FSSQL state/reward/Q implementation with masking removed; it is not claimed to reproduce an earlier unpublished scheduler exactly. Earlier energy-aware/multi-objective precursor manuscripts are not archival publications, so the revised novelty statement does not rely on them. Finally, we added a preregistered no-training Threshold controller based on temperature/backlog/risk rules.

Location in Revised Manuscript

Related Work and Positioning (including the new recent-work comparison table); Experimental Method–Baselines; Results; Supplemental Material.

Revised Text – Representative Excerpt

The revised paper does not claim novelty for shielding itself. Its contribution is the deployment-oriented combination of device-specific calibrated feasibility filtering, heterogeneous IDS model selection, complete online controller timing, and direct DC input-energy measurement.

Reviewer / AE Comment

The evidence that learning adds value over the shield alone is not convincing. Report dispersion, confidence intervals and paired significance tests; state-action-space size, learning/exploration schedules and a learning curve; discuss whether improvement justifies complexity.

Author Response

We replaced the previous aggregate comparison with matched-seed analysis. On Pi 3B+ (10 matched seeds), FSSQL-R is worse than Safe-Greedy: mean F1 0.8523 vs 0.8708, mean reward 0.5434 vs 0.5839, mean E2E latency 8.3149 vs 7.9775 ms, and switching cost 5.348 vs 0. Exact paired sign-flip tests give $p = 0.00195$ for F1/reward/switching and $p = 0.00586$ for E2E latency. Both methods have zero realized violations. On Pi 4B, five matched seeds per B0/B1/B2 condition show FSSQL-R F1 advantages of +0.1209/+0.1172/+0.1141 and reward advantages of +129.4/+124.8/+122.0. However, Safe-Greedy selects FISVDD in all 15 formal Pi 4B runs, with FPR=1.0000 and TN=0 throughout, so we retain it as the direct same-guard myopic reference but do not present it as a competitive detector selector. Paired t tests for the F1 differences are approximately 0.011–0.015 while exact Wilcoxon tests are 0.0625 because $n = 5$; we report both rather than choosing the more favorable test, and we separately compare FSSQL-R with the much stronger Threshold baseline. The revised implementation has 1,280 states and 5,120 state-action pairs, $\gamma = 0.95$, a

visit-count learning-rate schedule, epsilon decay from 1.0 to 0.05 by window 4,500, and a replay capacity of 512. At 600 windows FSSQL-R has visited 29.7 states/96.3 state-actions; at 6,000 windows 34.9/118.9. Greedy-policy change rate falls from 0.006875 to 0.000156. We therefore describe late-budget behavioral stability on visited states, not full-state mathematical convergence.

Location in Revised Manuscript

Method–Masked Q-learning; Results–RQ2 and Learning Stability; Supplemental Sections C–D.

Revised Text – Representative Excerpt

The learning results are not universal: FSSQL-R underperforms Safe-Greedy on the Pi 3B+ stationary binding regime, but improves F1 and reward relative to the same myopic reference on the Pi 4B. We therefore describe the incremental value of Q-learning as device- and regime-dependent.

Reviewer / AE Comment

The empirical setup does not demonstrate real deployment cost. Clarify measured vs reconstructed quantities, provide online Pi execution, measure controller/tail cost and switching, and validate physical energy.

Author Response

The revised principal deployment evidence is online, not replay. We executed full closed-loop scheduling on Pi 3B+ and Pi 4B 8GB. The Pi 4B formal matrix contains 60 accepted cells and 36,000 windows (4 policies $\times$ 3 background conditions $\times$ 5 matched seeds $\times$ 600 windows). Each window records preprocessing, telemetry, state construction, guard/admissible-set construction, policy selection, Q lookup, detector inference, switching, total controller time, full E2E latency, temperature, frequency and fault flags. Mean controller time is 3.919 ms for FSSQL-R and 3.885 ms for Safe-Greedy, compared with mean selected-detector inference of 0.415 and 0.426 ms. We also added direct POWER-Z KM003C measurements on the Pi 4B DC input path. Energy is integrated over the exact run boundary using timestamped samples. The energy results do not support a consistent FSSQL-R savings claim, and the manuscript states this explicitly.

Location in Revised Manuscript

Experimental Method–Online execution and Physical energy measurement; Results–RQ3; Fig. 1 and Fig. 2; Supplemental Sections E.

Revised Text – Representative Excerpt

Physical-energy results are reported as absolute KM003C DC input-side measurements. The measured differences do not establish a consistent FSSQL-R energy advantage.

Reviewer / AE Comment

Safety and generality claims should be more careful. Report residual distributions, calibration size and empirical coverage; test changed load/second device or scope claims. Also fix the table/figure/presentation issues.

Author Response

We agree and have removed unconditional “hard safety” wording from the empirical claims. Each device now has independent fit, margin and held-out coverage splits over 72 action/-workload/background cells, with 2,880 windows per split. Pi 3B+ joint held-out coverage is $2879/2880 = 0.999653$; Pi 4B joint coverage is 0.998958 . Misses are retained and no post-coverage tuning is performed. We also added a second hardware configuration (Pi 4B 8GB) and B0/B1/B2 background-load robustness. The Pi 3B+ binding workload has $|A_{safe}| = 3$ in every window and excludes LUCID; the Pi 4B online workloads have $|A_{safe}| = 4$, demonstrating hardware-dependent feasibility. The revised paper is explicitly scoped to Pi-class evidence and describes MCU applicability as prospective. The old Table VII/XIII/XIV and Figures 11/13 are replaced by newly generated tables/figures from the frozen analysis pipeline; a consolidated parameter table is added and the previously sparse discussion subsections are rewritten.

Location in Revised Manuscript

Tables I–V; Figs. 1–2; Discussion; Limitations.

Reviewer 1

Reviewer / AE Comment

1. Justify the practical advantage of Q-learning over Safe-Greedy; report dispersion and paired significance; examine switching cost.

Author Response

We agree that the earlier claim was not justified. The revised Pi 3B+ result is adverse to FSSQL-R and is retained: F1 0.8523 vs 0.8708; reward 0.5434 vs 0.5839; mean E2E 8.3149 vs 7.9775 ms; switching cost 5.348 vs 0; both zero realized violations. Exact paired sign-flip $p = 0.00195$ for F1/reward/switching. On Pi 4B, FSSQL-R has a consistent paired F1/reward advantage over Safe-Greedy across B0/B1/B2, but Safe-Greedy selects FISVDD in all 15 formal runs and has FPR=1.0000/TN=0 throughout. We retain it as the direct same-guard myopic reference. We also now report the seed-level FSSQL-R behavior rather

than only mean $\pm$ SD: seed 5102 repeatedly selects FISVDD in 97.0–98.7% of windows across B0/B1/B2 and yields FPR 0.9144/0.9780/0.9476, while the other four seeds average FPR 0.1384/0.1380/0.1517. Table V therefore reports FPR as median [min,max], and the supplement lists all five seed-level values. This directly changes the conclusion: Q-learning is not universally advantageous and its seed-level failure modes must be considered alongside operating regime.

Location in Revised Manuscript

Results–RQ2; Discussion–When is Q-learning worth its complexity?; Supplemental Sections C and E.

Reviewer / AE Comment

2. Clarify replay vs direct hardware execution; ideally provide an end-to-end online Pi experiment.

Author Response

The revised manuscript replaces replay-only deployment claims with online hardware execution. The Pi 3B+ binding campaign contains 36,000 online windows and the Pi 4B formal campaign contains 36,000 online windows. The Pi 4B chain executes input, preprocessing, telemetry, state construction, guard, policy, selected IDS, prediction and post-action scoring in real time. The manuscript now distinguishes action-level guard latency from complete E2E latency and reports controller timing explicitly.

Location in Revised Manuscript

Experimental Method–Baselines and online execution; Results–RQ1/RQ3.

Reviewer / AE Comment

3. The safety theorem relies on bounded prediction error, while margins are empirical and residual violations remain. Provide residual distributions, calibration size and empirical coverage.

Author Response

We agree and now present the theorem as conditional one-step admissibility. We report calibration sizes and held-out empirical coverage rather than treating Q0.99 margins as absolute guarantees. Pi 3B+ joint coverage is 0.999653 and Pi 4B is 0.998958. A retained Pi 3B+ Threshold tail event reaches 52.522 ms even though TinyDL was admitted with large predicted headroom; the event is dominated by a 50.414-ms controller/runtime outlier. This is used to show why complete-pipeline safety cannot be inferred solely from action-level residual margins.

Location in Revised Manuscript

Feasible-Set Method–Conditional safety statement; Table II; Results–RQ1; Limitations.

Reviewer / AE Comment

4. Strengthen generalization/robustness beyond a single Pi 3B+ or scope the claim.

Author Response

We added independent calibration and online validation on a Raspberry Pi 4B 8GB, plus B0/B1/B2 background CPU conditions. The devices show different feasible-set behavior: Pi 3B+ is binding ($|A_{safe}| = 3$, LUCID excluded), while Pi 4B remains nonbinding ($|A_{safe}| = 4$) in the formal online conditions. The revised manuscript scopes the empirical hardware claim to Pi-class devices and does not claim MCU validation.

Location in Revised Manuscript

Experimental Method–Hardware and calibration; Results–RQ1; Limitations.

Reviewer 2

Reviewer / AE Comment

1. Novelty relative to prior/closely related work is not sufficiently clear; clarify whether Unshielded-Q is the earlier scheduler and discuss shielded RL for IDS.

Author Response

The revised manuscript no longer presents shielding as the novelty. It positions the contribution as device-calibrated feasible-set enforcement for heterogeneous IDS model selection, paired with an explicit empirical decomposition of shield vs learning vs simple controllers. We cite Malwi *et al.* (CMC 2026, DOI 10.32604/cmc.2026.074274) and explain the difference from self-healing remediation. We also added recent 2024–2025 TSUSC papers on sustainable DRL scheduling/DVFS and IoT anomaly adaptation, recent TMC papers on workload-adaptive edge inference and DVFS/offloading, and recent TIFS papers on online/open-set IDS. The new comparison table shows that our distinctive combination is device-specific feasible-set calibration, heterogeneous IDS model selection, complete controller timing, and direct input-energy measurement. Unshielded-Q is now defined unambiguously as the same revised Q implementation with the mask removed. It is not represented as an exact historical implementation of earlier unpublished precursor work.

Location in Revised Manuscript

Related Work and Positioning.

Reviewer / AE Comment

2. Missing conventional threshold baseline.

Author Response

Added. The Threshold controller is no-training and uses frozen development-derived temperature, backlog and risk thresholds. Importantly, it is not a weak straw-man: on Pi 4B it has the highest mean F1 in B0/B1/B2 (0.9491/0.9533/0.9627) and mean FPR 0.0726/0.0581/0.0490, compared with FSSQL-R mean FPR 0.2936/0.3060/0.3109. The FSSQL-R distribution is strongly seed-dependent, so the main table reports its FPR as median [min,max] and the supplement gives all five seeds. The FSSQL-R-minus-Threshold paired FPR bootstrap 95% CIs are positive in all three conditions ([+0.0249,+0.5384], [+0.0323,+0.5928], [+0.0581,+0.5896]), while the paired F1-difference intervals cross zero. We therefore present Threshold as a strong practical baseline and explicitly avoid portraying FSSQL-R as the overall winner.

Location in Revised Manuscript

Experimental Method–Baselines; Table IV; Discussion.

Reviewer / AE Comment

3. Controller overhead is not adequately accounted for. Report mean/tail controller time and clarify whether controller/switching overhead enter accounting.

Author Response

All online windows now record controller decomposition and complete E2E latency. Across Pi 4B conditions, mean controller time is 3.919 ms for FSSQL-R and 3.885 ms for Safe-Greedy; detector inference is much smaller (about 0.415/0.426 ms). This confirms the reviewer’s concern that controller work is material. The manuscript separates guard-modeled action latency from complete E2E latency and reports switching separately.

Location in Revised Manuscript

Experimental Method–Online execution; Results–RQ3.

Reviewer / AE Comment

4. Energy/sustainability claim needs physical validation.

Author Response

We added direct inline DC input measurement using a POWER-Z KM003C. Each formal Pi 4B run is paired one-to-one with a timestamped meter trace, with at least 5 s pre/post capture and trapezoidal integration over the run ACK boundaries. We report W, J/run, J/window, J/1000 records and J/correctly detected attack in the artifact/supplement. The primary paper reports absolute J/run and explicitly states that the measured differences do not establish a consistent FSSQL-R physical-energy advantage.

Location in Revised Manuscript

Experimental Method–Physical energy measurement; Results–RQ3; Fig. 1; Supplemental Section E.

Reviewer / AE Comment

5. Q-learning convergence concerns: report state-action space, schedules and learning curve/stability.

Author Response

The revised implementation has 1280 states and 5120 state-action pairs. We report $\gamma = 0.95$, the visit-count alpha schedule, epsilon decay, replay capacity and update schedule. At 600 windows FSSQL-R visits 29.7 states/96.3 state-actions and has reward 0.483742; at 6000 windows 34.9/118.9 and reward 0.526027. The greedy-policy change rate drops from 0.006875 to 0.000156. We explicitly avoid a mathematical convergence claim over unvisited states.

Location in Revised Manuscript

Method–Masked Q-learning; Results–Learning stability; Supplemental Section D.

Reviewer / AE Comment

Other suggestion 1: Table VII should show the latency quantity actually compared against the cap.

Author Response

The old Table VII has been replaced. The revised method defines the guard quantity directly as predicted action/service latency plus the frozen residual margin, while complete E2E latency is separately measured online. The distinction is emphasized in the method and the retained controller-tail example.

Location in Revised Manuscript

Problem Formulation Eq. (3); Experimental Method; Results–RQ1.

Reviewer / AE Comment

Other suggestion 2: verify Table XIV because it did not match Table XI.

Author Response

The old tables are removed. Revised tables are generated from one fixed accepted-run manifest and analysis pipeline, preventing denominator/aggregation drift.

Location in Revised Manuscript

Revised calibration and policy tables and Supplemental Tables.

Reviewer / AE Comment

Other suggestion 3: provide a consolidated parameter table.

Author Response

Added as Table II (the new recent-work comparison is Table I).

Reviewer / AE Comment

Other suggestion 4: Figure 13 is empty and Figure 11 legend obscures curves.

Author Response

The old figures are replaced by new Pi 4B energy and background-load robustness figures generated from accepted runs. There is no empty-axis figure, and the revised robustness legend is positioned without obscuring the primary data.

Reviewer / AE Comment

Other suggestion 5: Section VI-F has no substantive content and VI-G is very brief.

Author Response

The result/discussion structure has been rewritten. The revised Results section contains RQ1–RQ4, followed by a Discussion section with dedicated subsections on shielding, Q-learning complexity, and sustainability.

Reviewer 3

Reviewer / AE Comment

1. Table XIII appears inconsistent with the definition of fallback and $\bar{\rho}_{safe}$.

Author Response

We agree. Rather than reinterpret the old inconsistent aggregate table, we removed it and rebuilt the safe-set reporting from window-level hardware data. In the accepted Pi 3B+ binding campaign, $|A_{safe}| = 3$ in all 36,000 windows and fallback=0. In the Pi 4B online campaign, $|A_{safe}| = 4$ in all 36,000 accepted windows and fallback=0. The revised paper reports these distributions directly, eliminating the denominator ambiguity.

Location in Revised Manuscript

Results–RQ1; Supplemental Sections B–C.

Reviewer / AE Comment

2. Clarify how replay evolves state and whether it is truly closed loop.

Author Response

The principal revised evidence is no longer replay. The online runner measures the hardware state and controller timing every window and executes the selected detector in the loop. We retain no claim that reconstructed trace replay itself proves closed-loop dynamics. A separately preregistered stateful queue extension was stopped before policy execution because its measured service-model precondition could not establish both backlog accumulation and recovery without changing the frozen design.

Location in Revised Manuscript

Experimental Method–Online execution; Discussion; Supplemental Section G.

Reviewer / AE Comment

3. Make the near-cap regime reproducible and report uncertainty; per-seed values/spread are needed.

Author Response

The revised Pi 3B+ binding workload is explicitly fixed and evaluated over 10 matched seeds, 600 windows each. The Pi 4B robustness conditions are fixed as n=16 with B0/B1/B2 background CPU conditions and five matched seeds. We report SDs, bootstrap CIs, paired tests and effect sizes. The additional per-seed results are provided in the supplementary

material.

Location in Revised Manuscript

Consolidated parameter table; Results policy tables; Supplemental Sections C and E.

Reviewer / AE Comment

4. Explain why FSSQL-R had fewer realized violations than Safe-Greedy despite sharing the same admissible set.

Author Response

The newly rebuilt hardware evidence does not reproduce the old 0.1-vs-1.8 gap, and we do not preserve that unsupported interpretation. In the Pi 3B+ binding regime, both FSSQL-R and Safe-Greedy have zero realized violations. The only retained Pi 3B+ latency violation is from Threshold and is traced to a controller/runtime tail event, not a detector selected outside the guard. This revision therefore removes the earlier causal claim that learning itself reduced residual violations.

Location in Revised Manuscript

Results–RQ1/RQ2; Discussion.

Revised Text – Representative Excerpt

The rebuilt online evidence does not reproduce the earlier 0.1-versus-1.8 residual-violation gap. Both FSSQL-R and Safe-Greedy have zero realized violations in the accepted Pi 3B+ binding campaign, so the earlier causal interpretation is removed.

Reviewer / AE Comment

5. Soften “shielding is necessary” and scope MCU claims.

Author Response

Done. The manuscript states that shielding provides calibrated feasible-set enforcement when constraints become binding, not that all safe operation requires a shield. Static/threshold policies can be safe without the shield. Hardware claims are explicitly scoped to Raspberry Pi-class devices; MCU applicability is prospective only.

Location in Revised Manuscript

Abstract; Introduction; Discussion; Limitations.

Reviewer / AE Comment

6. Complete positioning against closely related work; cite earlier energy-aware/multi-objective papers if public or clarify status.

Author Response

We clarify that the earlier precursor formulations referenced in the submitted discussion are not archival publications and are not used as evidence of novelty. Unshielded-Q is defined solely as a transparent revised baseline. We additionally cite the recent published shielded-RL IDS paper requested by Reviewer 2 and distinguish its action/safety setting from ours. To complete the positioning more broadly, the revision also adds recent 2024–2025 TSUSC work on sustainable runtime scheduling and IoT anomaly adaptation, TMC work on adaptive edge inference/DVFS, and TIFS work on online and lightweight dynamic IDS.

Location in Revised Manuscript

Related Work and Positioning.

Summary of Revision Evidence

For clarity, the core new evidence is summarized below.

Item	Revised evidence
Pi 3B+ guard	2,880-window held-out coverage; joint $2879/2880 = 0.999653$; no post-coverage tuning.
Pi 3B+ binding comparison	6 policies, 10 matched seeds, 36,000 online windows; $ A_{safe} = 3$ throughout.
Pi 4B guard	Independent fit/margin/coverage; joint held-out coverage 0.998958.
Pi 4B online comparison	4 policies $\times$ 3 loads $\times$ 5 matched seeds $\times$ 600 windows = 36,000 accepted online windows.
Physical energy	60 accepted POWER-Z KM003C traces bound to online runs; absolute DC input energy.
Official test	One-time 82,332-row evaluation with fixed preprocessing and thresholds after all development choices were finalized.

We appreciate the reviewers’ comments, which led us to replace several over-strong interpretations with measured, device-specific conclusions and to provide substantially stronger deployment evidence.

Final Notes on Formatting and Submission

Author Response

We have carefully revised the manuscript in response to the Associate Editor and all three reviewers. The revised submission now separates calibration evidence, online hardware evidence, physical energy measurements, and the one-shot detector-generalization test; it also reports adverse results where they occur rather than preserving earlier replay interpretations. All principal policy comparisons use seed-level statistical units, and the revised paper explicitly distinguishes empirical guard coverage from a universal end-to-end

safety guarantee.

The revised main manuscript is submitted in IEEE double-column format together with the Supplementary Material and this point-by-point response letter. The manuscript source has been checked for unresolved references, draft markers, incomplete placeholders, and presentation defects. We respectfully submit the revision for further consideration by the IEEE Transactions on Sustainable Computing.

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